EXPERIMENT 5

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Experiment 5 – SQL Conditional Logic (Odd & Even Values)

SQL Conditional Logic using the MOD (%) operator to classify employee salary values as Odd or Even. This experiment demonstrates numerical analysis and conditional evaluation using SQL queries in Oracle and PostgreSQL.

Aim

To understand and apply conditional logic in SQL by using the modulus operator (MOD / %) to analyze numerical data and classify employee salaries as odd or even, thereby improving data analysis and decision-making skills in SQL.

Objective

- To write and execute SQL queries using the MOD (%) operator.
- To retrieve salary details from an employee table.
- To classify salaries into Odd and Even categories.
- To display odd and even salary values separately.

Software Requirements

Database Management System:
Oracle Database Express Edition (Oracle XE)
PostgreSQL Database

Database Administration Tool / Client Tool:
Oracle SQL Developer (for Oracle XE)
pgAdmin (for PostgreSQL)

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Practical / Experiment Steps

1. Create an employee table with columns:
   id
   name
   department
   salary
2. Insert sample employee salary records into the table.
3. Write a SQL query to retrieve employee salary details.
4. Use the MOD(salary, 2) function (Oracle) or salary % 2 (PostgreSQL) to check whether the salary is odd or even.
5. Use a CASE statement to classify salaries as:
   'Odd Salary'
   'Even Salary'
6. Execute separate queries to:
   Display only employees with Odd salaries.
   Display only employees with Even salaries.
7. Observe and record the output results.

Procedure

1. Open Oracle SQL Developer or pgAdmin.
2. Connect to the database.
3. Create the employee table using CREATE TABLE.
4. Insert sample records using INSERT INTO.
5. Execute SELECT queries with the MOD operator.
6. Apply CASE statement for conditional classification.
7. Display results and capture screenshots.

Input / Output Details

Input

Table Name: employee

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Columns:
id (Integer)
name (Varchar)
department (Varchar)
salary (Integer)

Logic Used:
MOD(salary, 2) or salary % 2
CASE statement

Code

```
CREATE TABLE employee (
    emp_id INT PRIMARY KEY,
    emp_name VARCHAR(100),
    salary INT
);

INSERT INTO EMPLOYEE VALUES (101, 'Siya', 25000);
INSERT INTO EMPLOYEE VALUES (102, 'Rohit', 45000);
INSERT INTO EMPLOYEE VALUES (103, 'Ali', 28999);
INSERT INTO EMPLOYEE VALUES (104, 'Piya', 32001);
INSERT INTO EMPLOYEE VALUES (105, 'Bheem', 50000);

SELECT * FROM EMPLOYEE;

SELECT * FROM EMPLOYEE
WHERE SALARY % 2 = 0;

SELECT * FROM EMPLOYEE
```

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```
WHERE SALARY % 2 <> 0;


SELECT EMP_ID,
      EMP_NAME,
      SALARY,
      CASE
         WHEN SALARY % 2 = 0 THEN 'Even Salary'
         ELSE 'Odd Salary'
      END AS SALARY_TYPE
FROM EMPLOYEE;
```

Step-wise Output

Step 1 – Create employee Table

```
CREATE TABLE employee (
    emp_id INT PRIMARY KEY,
    emp_name VARCHAR(100),
    salary INT
);
```

Data Output Messages Notifications

```
CREATE TABLE

Query returned successfully in 78 msec.
```

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Step 2 – Insert Data into employee Table

```
INSERT INTO EMPLOYEE VALUES (101, 'Siya', 25000);
INSERT INTO EMPLOYEE VALUES (102, 'Rohit', 45000);
INSERT INTO EMPLOYEE VALUES (103, 'Ali', 28999);
INSERT INTO EMPLOYEE VALUES (104, 'Piya', 32001);
INSERT INTO EMPLOYEE VALUES (105, 'Bheem', 50000);
```

Data Output | Messages | Notifications

```
INSERT 0 1

Query returned successfully in 57 msec.
```

Step 3 – Display Employee Salary Records

```
SELECT * FROM EMPLOYEE;
```

Data Output | Messages | Notifications

	emp_id [PK] integer	emp_name character varying (100)	salary integer
1	101	Siya	25000
2	102	Rohit	45000
3	103	Ali	28999
4	104	Piya	32001
5	105	Bheem	50000

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Step 4 – Classify Salaries as Odd or Even

```
SELECT * FROM EMPLOYEE
WHERE SALARY % 2 = 0;
```

Data Output Messages Notifications

	emp_id [PK] integer	emp_name character varying (100)	salary integer
1	101	Siya	25000
2	102	Rohit	45000
3	105	Bheem	50000

Step 5 – Display Only Odd Salaries

```
SELECT * FROM EMPLOYEE
WHERE SALARY % 2 <> 0;

SELECT EMP_ID,
```

Data Output Messages Notifications

	emp_id [PK] integer	emp_name character varying (100)	salary integer
1	103	Ali	28999
2	104	Piya	32001

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Step 6 – Display Only Even Salaries

```sql
SELECT * FROM EMPLOYEE
WHERE SALARY % 2 = 0;
```

Data Output Messages Notifications

	emp_id [PK] integer	emp_name character varying (100)	salary integer
1	101	Siya	25000
2	102	Rohit	45000
3	105	Bheem	50000

```sql
SELECT EMP_ID,
       EMP_NAME,
       SALARY,
       CASE
           WHEN SALARY % 2 = 0 THEN 'Even Salary'
           ELSE 'Odd Salary'
       END AS SALARY_TYPE
FROM EMPLOYEE;
```

Data Output Messages Notifications

	emp_id [PK] integer	emp_name character varying (100)	salary integer	salary_type text
1	101	Siya	25000	Even Salary
2	102	Rohit	45000	Even Salary
3	103	Ali	28999	Odd Salary
4	104	Piya	32001	Odd Salary
5	105	Bheem	50000	Even Salary

Learning Outcome

After completing this experiment, the learner will be able to:

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- Understand conditional logic in SQL queries.
- Use the MOD (%) operator for numerical analysis.
- Classify numeric values into odd and even categories.
- Write efficient SQL queries for payroll data analysis.
- Apply SQL logic in real-world enterprise environments.